

HANS YANG

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Education

Wilfrid Laurier University

Sep. 2024 – Present

Bachelor of Science (Computer Science)

Major GPA: 11.89/12 (89%)

Relevant Courses: Algorithm Design & Analysis, Calculus I & II, Computer Networks, Databases I, Software Engineering, Data Structures I & II, Introduction to Microprocessors, Discrete Mathematics, Object Oriented Programming, Probability I

Technical Skills

Languages: C, C++, HTML/CSS, Go, Java, JavaScript, Lua, Python, R, SQL, TypeScript, VBA

Software & Tech: AWS Lambda, AWS S3, Azure, Docker, Excel/Google Sheets, Git, GitHub, Matplotlib, MongoDB, MySQL, Node.js, NumPy, Pandas, Redis, Vercel, Vite

Projects

🔗 **AI Flashcard Study Tool** | *AWS Lambda, AWS S3, Docker, Gemini API, Go, Node.js, Typescript* 2026 – Current

- Designing a full-stack serverless AI document-processing pipeline using **AWS Lambda**, **AWS S3**, and **Docker** to transform uploaded files into structured flashcard data through automated text extraction and generation workflows.
- Developing backend services with **Go** and client integrations with **TypeScript/Node.js**, establishing an end-to-end architecture for document ingestion, AI processing, and user-facing content delivery.

🔗 **Visulie** | *Lua, Roblox Studio, State/Data-Driven Architecture* 2021 – Current

- Engineered a Roblox Marketplace animation plugin with a custom editor, serialization pipeline, and runtime interpreter for real-time multi-property animation playback, reducing manual scripting effort by up to 90% and generating **150+ sales and \$1.5K+ CAD revenue**.
- Actively ranked #1 on Google Search**, increasing plugin visibility and outreach through marketplace optimization and community engagement.
- Iterated on a customer-facing product, leveraging user feedback, feature requests, and bug reports to drive continuous improvements and deliver a more robust user experience.

🔗 **FPS** | *Lua, Roblox Studio* 2023

- Shipped a simulator-style progression game built around upgrade-driven gameplay loops and progression systems, scaling to **100K+ total visits** and **100+ peak CCU** through iterative gameplay system tuning, economy balancing, and UX-driven front-end design.
- Featured in the official Roblox Developer 2022 Year-in-Review** for applying Parallel Lua and Canvas Groups with awareness of parallel execution constraints to improve performance and rendering throughput.

🔗 **Pocket Rift** | *Lua, Roblox Studio, State Machine/Event/Authoritative Server Architecture* 2019 – Current

- Implemented user tracking through economy and funnel analytics to measure onboarding behavior, enabling iterative optimizations that improved conversion rates by over **185%**.
- Developed a multiplayer networking architecture with client-side prediction, snapshot-based state replication, and delta compression, reducing latency and **decreasing network load by up to 95% over baseline implementations**.
- Engineered reusable engine frameworks such as a **Redis-inspired cache for player sessions** and **React-inspired reactive state management**, reducing coupling between systems and enabling scalable client-server architecture.

🔗 **SPY Trading Algorithm** | *Python, Matplotlib, yfinance* Jan. 2026

- Designed and implemented a Python-based systematic trading strategy for SPY, backtested on historical market data with performance analytics, **outperforming buy-and-hold by approximately 54%**.
- Analyzed performance differences between a rule-based trading approach and buy-and-hold, identifying limitations such as sensitivity to market conditions and transaction assumptions.

Work Experience

Revlayz Software Development

2026 (40h/Week)

Fullstack Gameplay Programmer, CI/CD

Virtual

- Engineered backend systems for player data persistence and implemented behavior trees for NPC decision-making, delivering new gameplay features coinciding with a **peak of 10,000+ concurrent users**.
- Developed frontend UI and VFX, contributing to the player-facing experience for a title that **scaled to 10K+ CCU**.
- Sourced, vetted, and onboarded external contractors, negotiating pricing and scope to align deliverables with project needs and budget.
- Shipped content under tight deadlines with unit and integration test coverage, **boosting D1 retention by over 10%**.
- Leveraged AI to improve documentation and navigate a complex legacy codebase, creating a more productive and maintainable development environment.